

# Tree growth responses to extreme drought events are not well predicted by climate

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## OVERVIEW

- Climate change is **intensifying droughts** through shifts in temperature, vapor pressure deficit (VPD), and soil moisture
  - Combined effects of these drivers on tree growth remain poorly understood at large scales
  - A new **Bayesian approach** to jointly model tree growth across Western US, at the tree level
- Our model is able to separate climate effects from other complex dynamics (e.g. allometry)
  - Some extreme drought caused **exceptional growth reductions** far exceeding what standard climate variables predict
  - Current models likely **underestimate growth declines** during extreme events

## 1. BACKGROUND

- Tree-ring width time series are one of the longest and most spatially extensive records of forest response to climate
- However, tree rings are affected by many drivers other than climate: age-size effects, competition, biotic disturbance...
- Thus, standard dendrochronology approaches detrend individual trees and aggregate them to identify some shared climatic signal
- But drought stress operates at the individual level, whose responses integrate non-linear physiological processes
- To capture how extreme events, we need to model individual trees explicitly



## 2. MODELING GROWTH AT THE TREE LEVEL

|                                                                                |                                                               |
|--------------------------------------------------------------------------------|---------------------------------------------------------------|
| Growth is resulting from three major components (here show a non-shocky tree?) |                                                               |
| Long-term trajectories....<br><br>FIGURE                                       | Short- to mid-term dynamics....<br><br>FIGURE                 |
| Climate : 3 predictors<br><br>FIGURE                                           | All together: (minus shock)<br><br>Predictions at tree-level! |

## 3. MAIN CLIMATIC DRIVERS OF GROWTH

Some stuff about the predictors, how much variation they explain...

FIGURE OF CLIMATE PRED IMPACT ACROSS WESTERN US

Highlight some good and bad periods?

## 4. EXTREME GROWTH REDUCTIONS

Some individual trees have extreme responses

FIGURE : a shocky tree, with predictiosn fro model without shocks?

These extreme are often experienced by several trees I na forest

FIGURE : shocky tree 2

FIGURE : shocky tree 3

FIGURE : shocky tree 3

FIGURE : shocky tree 4

These extreme have major consequences at ecosssystem scale

FIGURE OF SHOCKS ACROSS WESTERN US

Blablabla about shocks  
% of variation explained  
  
Major droughts

## 5. IMPLICATIONS FOR FORESTS AND FORECASTS

References...